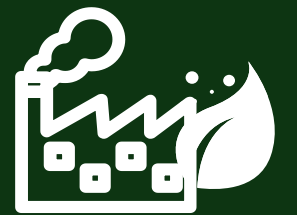




AI-BASED GREENWASHING DETECTION IN CORPORATE SUSTAINABILITY REPORTS



AI, Power & Responsibility – Governing Intelligent Systems for Sustainability



1. MOTIVATION

Greenwashing undermines trust in Sustainability reporting and misleads Investors and stakeholders.

Companies may exaggerate or use vague language in ESG disclosures Without sufficient evidence

Responsible AI can help identify Misleading claims and support Transparency and accountability



2. DATASETS



S&P 500 SUSTAINABILITY REPORTS Dataset

- 5,180 sustainability reports
- 500 companies
- 2011-2021



S&P 500 ESG Risk Ratings Dataset

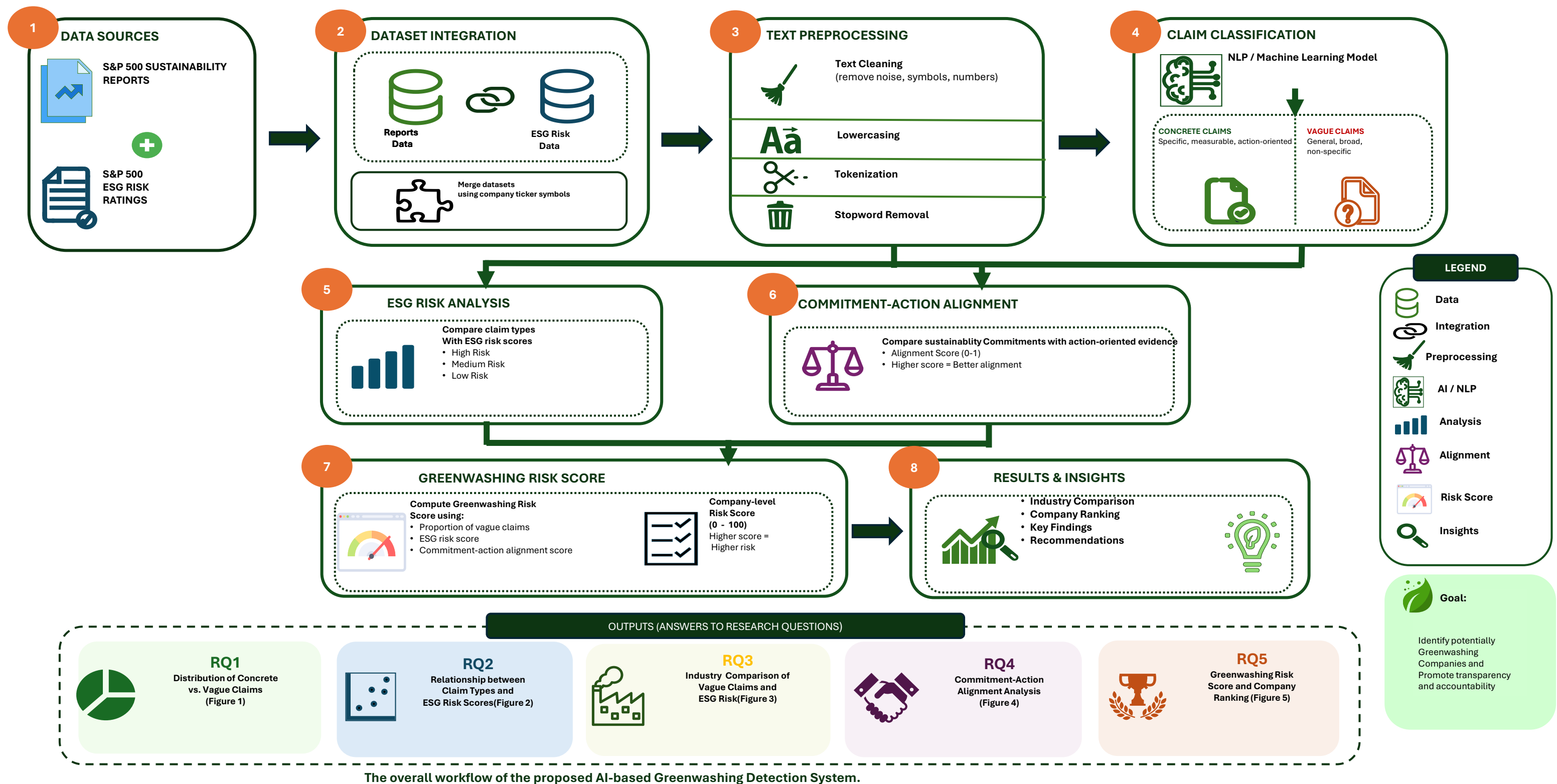
- ESG risk scores and percentile rankings
- Provided by Sustainalytics



3. RESEARCH QUESTIONS

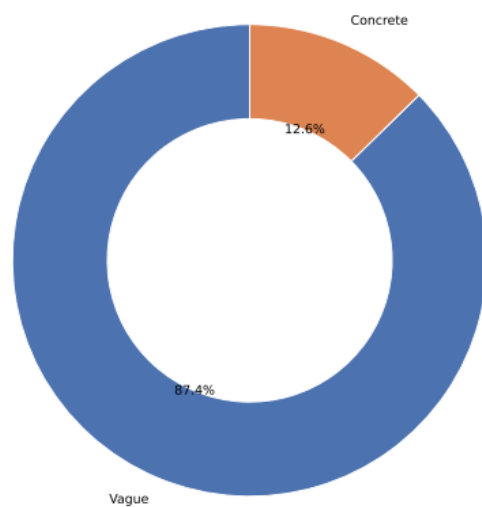
- RQ1 What proportion of sustainability report claims are vague versus concrete?
- RQ2 How do different claim types relate to ESG risk score?
- RQ3 Which industries show the highest proportion of vague sustainability claims?
- RQ4 Do companies align their commitments With action-oriented evidence?
- RQ5 Which companies exhibit the highest potential greenwashing risk?

4. METHODOLOGY WORKFLOW

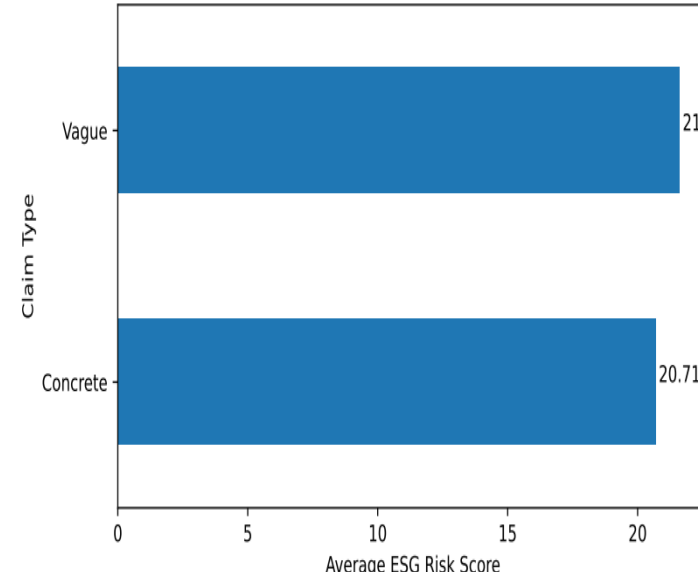


5. RESULTS (KEY FINDINGS)

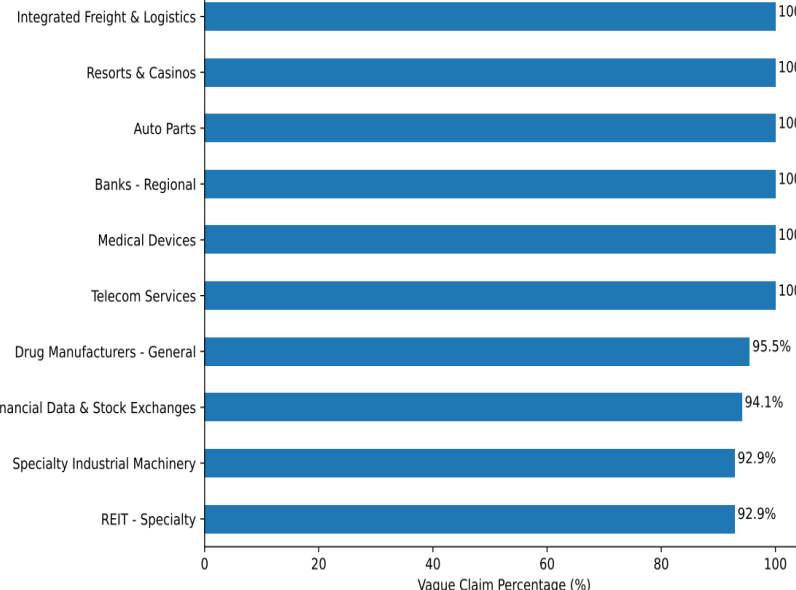
Distribution of Claim Types in Sustainability Reports



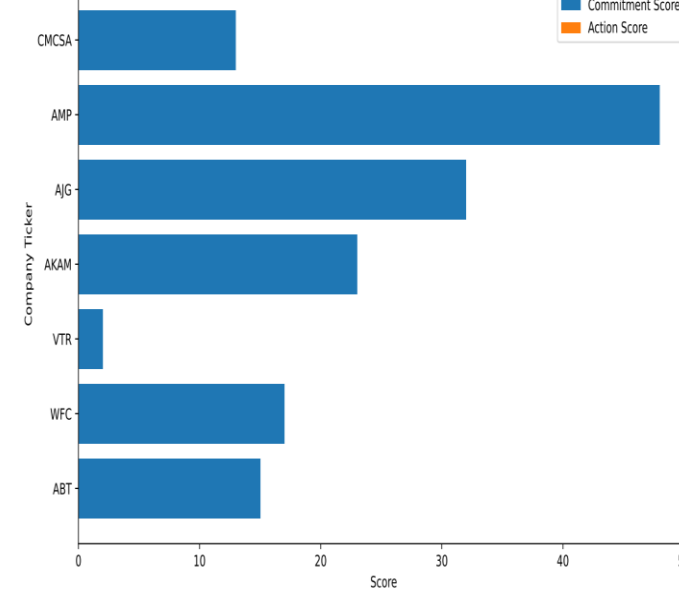
Average ESG Risk Score by Claim Type



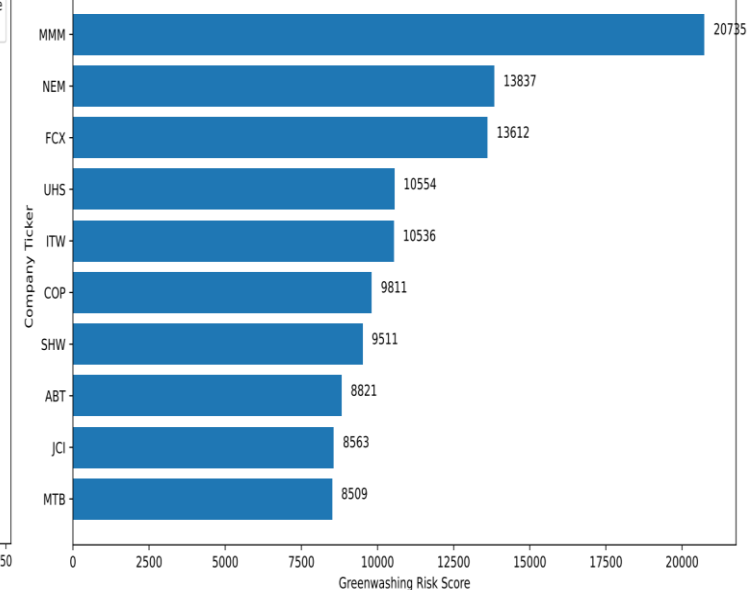
Top Industries by Vague Claim Percentage



Commitment vs Action Scores for Lowest Alignment Companies



Top Companies by Greenwashing Risk Score



6. SUMMARY OF KEY FINDINGS

- More vague claims than concrete claims were observed.
- ESG risk levels differed across industries.
- Some companies showed weak commitment-action alignment.
- The Greenwashing Risk Score identified higher-risk companies.
- Text analysis and ESG data improved company assessment

7. CONTRIBUTIONS

- AI-based greenwashing detection framework.
- Integration of sustainability reports and ESG ratings.
- Commitment-action alignment analysis.
- Company-level Greenwashing Risk Score.
- Visual analysis through figures and tables

8. CONCLUSION

An AI-based framework was developed to detect potential greenwashing in corporate sustainability reports. The results showed differences in ESG risk and commitment-action alignment across companies. The proposed Greenwashing Risk Score helped identify companies with higher greenwashing risk and supported more transparent sustainability reporting.